

ΑΡΙΣΤΟΤΕΛΕΙΟ ΠΑΝΕΠΙΣΤΗΜΙΟ ΘΕΣΣΑΛΟΝΙΚΗΣ
ΣΧΟΛΗ ΘΕΤΙΚΩΝ ΕΠΙΣΤΗΜΩΝ
ΤΜΗΜΑ ΜΑΘΗΜΑΤΙΚΩΝ



A study of the Euclidean attack on RSA

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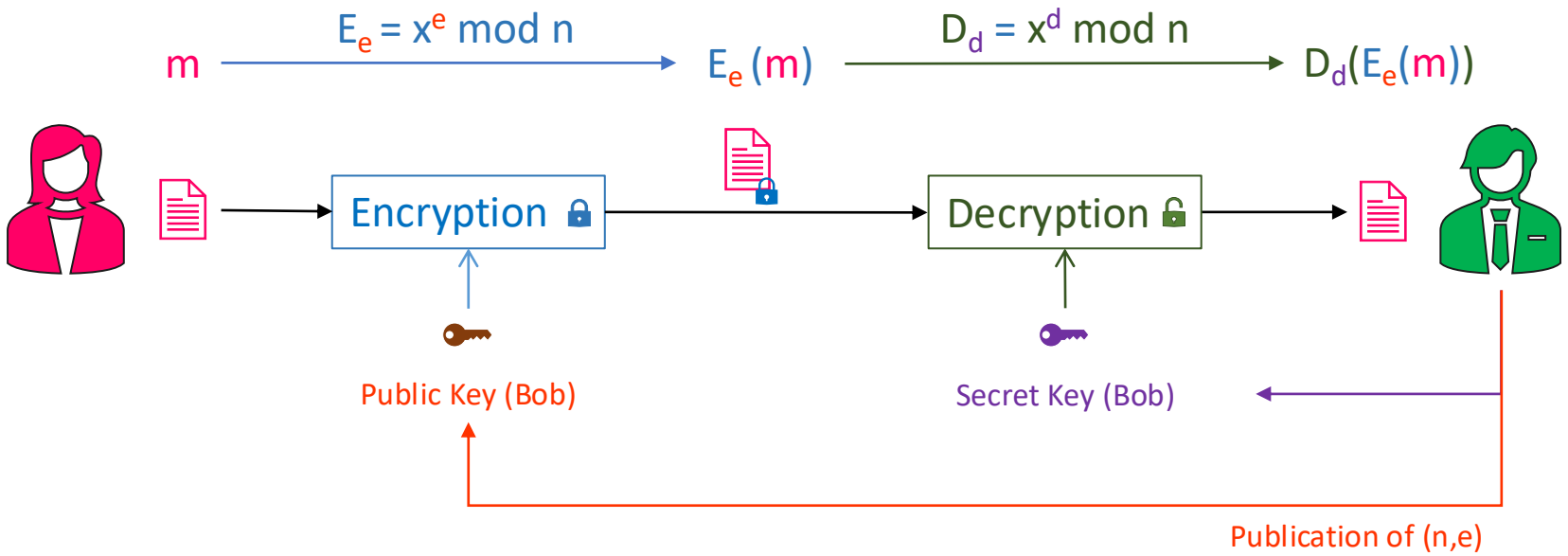
RSA Cryptosystem

Primes, > 2 of the same size

$n = p \cdot q$

$K = \{k \in \mathbb{Z} \mid 1 < k < \varphi(n) \ \& \ \gcd(k, \varphi(n))\}$

$ed \equiv 1 \pmod{\varphi(n)}$



Small exponent attacks on RSA

1. M.J. Wiener. “Cryptanalysis of short RSA secret exponents” (1990)
   -> $d < n^{1/4}/3$ & $O((logn)^3)$

2. Andrej Dujella. “Continued fractions and RSA with small secret exponent” (2004)
3. Eric R. Verheul and Henk C. A. van Tilborg. “Cryptanalysis of ‘Less Short’ RSA Secret Exponents” (1997)
   -> Slightly bigger d

4. Alexander May. “Using LLL-Reduction for Solving RSA and Factorization Problems” (2010)
   -> Wiener’s in $O((logn)^2)$

5. Johannes Blömer and Alexander May. “Low Secret Exponent RSA Revisited” (2001)
6. Dan Boneh and Glenn Durfee. “Cryptanalysis of RSA with private key d less than N0.292” (2000)
   -> $d < n^{0.292}$

Wiener's Attack

$$ed \equiv 1 \pmod{\varphi(n)} \Rightarrow \underbrace{\frac{e}{pq}} = \frac{k}{dg}(1-\delta)$$

public information
&
close underestimate of δ

What about (very) large exponents ?

7. M. Jason Hinek “(Very) Large RSA Private Exponent Vulnerabilities ” (2004)

   - Wiener’s attack works for $\sqrt{6}(\varphi(n) - d) < n^{1/4}$. *
   - If [5], [6] work for all $d < n^{\delta}$, then they work for all $d > \varphi(n) - n^{\delta}$.

     $d < n^{0.292}$

*$d > \varphi(n) - {n^{1/4}}/{\sqrt{6}}$

Euclidean Attack on RSA



Preliminaries

- p & q: two primes, > 2 of the same size and set $n = p \cdot q$
- $e, d \in (1, \varphi(n))$ such that $ed \equiv 1 \pmod{\varphi(n)}$
- $a = n + 1 \; mod \; e$
- $\Delta = \gcd(e, a)$

We set $r_0 = e$ and $r_1 = a$, then there are pairs of integers (q_i, r_{i+1}) $(i = 1, \ldots, m)$ such that $r_m = \Delta$, $r_{m+1} = 0$ and

$$r_{i-1} = r_i q_i + r_{i-1}, \; 0 < r_{i+1} < r_i.$$

Furthermore, we define integers s_i and t_i for $i = 1, \ldots, m+1$ as follows:

$$s_0 = 1, s_1 = 0 \text{ and } t_0 = 0, t_1 = 1$$

and for $i = 1, \ldots, m$,

$$s_{i+1} = s_{i-1} - s_i q_i \text{ and } t_{i+1} = t_{i-1} - t_i q_i.$$

It can be proven [9] that those integers t_i and s_i satisfy $|t_i| < e/r_{i-1}$, $|s_i| < a/r_{i-1}$ and

$$es_i + at_i = r_i \; (i = 0, \ldots, m+1).$$

- $\mu_i = \gcd(t_i, r_i)$ with $t'_i = t_i/\mu_i$ & $r'_i = r_i/\mu_i$ $(i = 0, \ldots, m+1)$

Main Theorem

ed ≡ 1 (mod φ(n))

Theorem 1. Let $e > n/c$, where c is an integer ≥ 2 and $k = (ed-1)/\phi(n)$. Let j be an index such that r_j is the largest remainder among them which are $< e^{3/4}$. The results below follow:

(i) Suppose that k is $\leq e^{1/4}/6\sqrt{c}$. Then, we have $\Delta < e^{3/4}$ and

$$k = |t'_j|, \quad p + q = (a + |t'_j|^{-1}) \, mod \, e.$$

(ii) Suppose that $e - k$ is $\leq e^{1/4}/6\sqrt{c}$. Then, we have $\Delta < e^{3/4}$ and

$$k = |t'_j|, \quad p + q = (a + (e - |t'_j|)^{-1}) \, mod \, e.$$

*$a = n + 1 \, mod \, e$
*$\Delta = \gcd(e, a)$
*$\mu_i = \gcd(t_i, r_i)$ & $t'_i = t_i/\mu_i \;\; (i = 0, \ldots, m+1)$

EUCLID-ATTACK Algorithm

* EUCLID-ATTACK algorithm *

· Input: An RSA public key (n, e) with $e > n / c$

· Output: The primes p and q (or FAIL)

Time Complexity: $O((loge)^2)$

1. Compute $a = (n+1) \, mod \, e$
2. Using the extended Euclidean algorithm for e and a, compute the biggest remainder r_j among them which are $< e^{3/4}$, as well as the associated integers s_j, t_j such that $s_j e + a t_j = r_j$.
3. Compute $\mu_j = gcd(t_j, r_j)$ and next $t'_j = t_j / \mu_j$.
4. Compute $\beta_1 = (a + |t'_j|^{-1}) \, mod \, e$ and next the solutions u_1 and v_1 of equation $x^2 - \beta_1 x + n = 0$. If u_1 and v_1 are positive integers, then output (u_1, v_1). Otherwise, go to the next step.
5. Compute $\beta_2 = (a + (e - |t'_j|)^{-1}) \, mod \, e$ and next the solutions u_2 and v_2 of equation $x^2 - \beta_2 x + n = 0$. If u_2 and v_2 are positive integers, then output (u_2, v_2). Otherwise, output FAIL.



Python Implementation:

EUCLID-ATTACK